Course Name: Geographic Information System

Week 11 - Assignment 11 - (Jul-2026)

TYPE OF QUESTIONS: MCQ/MSQ

Number of questions: 10 **Total marks: 10x1= 10**

QUESTION 1

Which command is used to load a package in R?

a) import(raster)
b) library(raster)
c) load(raster)
d) package(raster)

Answer: (b)

Detailed Solution:

The library () function is used in R to load installed packages into the current session.

QUESTION 2

Which function is used to create a dataframe in R?

A) dataframe()
B) create.frame()
C) data.frame()
D) make.table()

Answer: (c)

Detailed Solution:

data.frame() is the standard function in R for creating tabular datasets.

QUESTION 3

Which function is commonly used to display False Colour Composite (FCC) images in R?

a) plotFCC()
b) plotRGB()
c) fccPlot()

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d) rasterRGB()

Answer: (b)

Detailed Solution:

plotRGB() is used for plotting multi-band raster images as FCC or true-color composites.

QUESTION 4

The cellStats() function in R is mainly used to calculate _____.

a) Raster projection
b) Raster statistics
c) Vector topology
d) Attribute joins

Answer: (b)

Detailed Solution:

cellStats() computes raster statistics such as minimum, maximum, mean, and standard deviation.

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QUESTION 5

Data created in Google Earth can be imported into QGIS using which format?

a) TIFF
b) JPEG
c) KML
d) PNG

Answer: (c)

Detailed Solution:

KML is the standard format used for transferring spatial data from Google Earth into GIS software.

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QUESTION 6

Which dataset is freely available in the Indian Space Research Organisation Bhuvan Geoportal?

a) QuickBird
b) IKONOS
c) LISS III
d) WorldView

Answer: (c)

Detailed Solution:

LISS III satellite imagery is freely accessible through the Bhuvan Geoportal.

QUESTION 7

Which interpolation method uses 16 neighboring pixels?

a) Bilinear interpolation
b) Nearest neighbour
c) Bicubic interpolation
d) Linear interpolation

Answer: (c)

Detailed Solution:

Bicubic interpolation uses the nearest 16 pixels to create smoother resampled images.

QUESTION 8

The process of assigning real-world coordinates to a raster image is called _____.

a) Vectorization
b) Classification
c) Georeferencing
d) Rasterization

Answer: (c)

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Detailed Solution:

Georeferencing links raster images with actual geographic coordinates on Earth.

QUESTION 9

Which GIS operation creates zones around spatial features?

a) Overlay
b) Buffer
c) Classification
d) Interpolation

Answer: (b)

Detailed Solution:

Buffering creates polygon zones at specified distances around spatial features.

QUESTION 10

Which GIS software is free and open-source?

a) ArcGIS
b) ENVI
c) ERDAS Imagine
d) QGIS

Answer: (d)

Detailed Solution:

QGIS is an open-source GIS software used for mapping and spatial analysis.

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